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Search within or beyond the industrial cluster? The effect of perceived competition and knowledge base tacitness on strategic location choices of external knowledge search

Qiang Li ^a and Jingjing Guo^{b,c}

^aSchool of Business, Zhejiang University City College, Hangzhou, People's Republic of China; ^bInstitutes of Science and Development, Chinese Academy of Sciences, Beijing, People's Republic of China; ^cSchool of Public Policy and Management, University of Chinese Academy of Sciences, Beijing, People's Republic of China

ABSTRACT

From a knowledge-based perspective, this paper highlights the need for a framework linking perceived competition of the industry and firms' strategic location choices of external knowledge search and examines whether the perceived competition increases the propensity of cluster firms to search more widely from extra-cluster knowledge sources than intra-cluster knowledge sources. Furthermore, we emphasise that cluster firms with varying degrees of tacitness in their knowledge base might respond to such competition differently when conducting external knowledge search activities. Using a sample of 310 cluster firms in the Zhejiang Province of China, we find that the cluster firm would increase the propensity for more geographic boundary-spanning search relative to local search while under the pressure of competition. Moreover, the positive relationship between perceived competition and the propensity towards geographic knowledge search is weaker when the tacitness of the cluster firm's knowledge base is higher. The findings contribute to the understanding of the relationship between competition and the choice of location strategies in external knowledge search.

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Competition; geographic boundary-spanning search; knowledge base tacitness; industrial cluster

1. Introduction

As the knowledge-based view (KBV) argues, the heterogeneous knowledge base among firms is the main determinant of the creative capacity and above-average performance because it influences a firm's opportunities to reduce costs or increase differentiation from competitors (Grant 1996; Maskell and Malmberg 1999; del-Corte-Lora, Molina-Morales, and Vallet-Bellmunt 2016; Chen, Zhou, and Probert 2017). When a firm confronts a severe competition, it may be triggered to search for new ideas and knowledge which may help to reconfigure and update its knowledge portfolios to defend the existing competitive advantage (Chen 2008; Theeke 2016). The innovation search pieces of literatures suggest that it is important to engage with a wide range of external sources and actors, such as competitors, suppliers, clients, and universities, for acquiring new ideas, insight and knowledge to stay abreast of the competition (Laursen and Salter 2006; del-Corte-Lora, Molina-Morales, and Vallet-Bellmunt 2016). However, the nature of the relationship between competition and external knowledge search is not clear-cut. On the one hand, scholars state that a firm may be expected to pursue more external knowledge search if it will be able to obtain new and

CONTACT Qiang Li  liq@zucc.edu.cn, liq312@hotmail.com  School of Business, Zhejiang University City College, Hangzhou 310015, People's Republic of China

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heterogeneous knowledge to differentiate the products or lower production costs in reaction to the competition (Menon and Pfeffer 2003). On the other hand, competition pushes the firm to be more efficient and inhibits the pursuing of risky activities (Melville, Gurbaxani, and Kraemer 2007). For example, Morandi Stagni, Fosfuri, and Santalo (2017) find that tougher product-market competition causes an increase in technological exploitation and a decrease in technological exploration. Thus, competition may promote more local, less risky and costly search (Chen and Miller 2007; García-Granero, Vega-Jurado, and Alegre 2014). One of the possible explanation for the inconsistency is that these studies fail to distinguish the type of external knowledge search with different geographic location choices.

The literature in innovation search and economic geography has a debate over the differences of local search and geographic boundary-spanning search (Bathelt, Malmberg, and Maskell 2004; Alcácer and Chung 2007). It emphasises the relative advantage of local search in the efficiency of knowledge search process due to the geographic and social proximity of spillover (Letaifa and Rabeau 2013; Roper, Love, and Bonner 2017). Conversely, the relative advantages of geographic boundary-spanning search against local search are from increasing the probability of acquiring heterogeneous knowledge and reducing the likelihood of leakage of its own knowledge (Wang 2015). Given these differences, a firm would strategically choose the locations of search to trade-off between attaining the efficiency of the knowledge search process, and acquiring heterogeneous knowledge and reducing knowledge leak-out. Furthermore, under fierce competition, the managerial decisions of a firm are not only to increase or decrease the magnitude of external search but more importantly to make choices about how much to invest in local search and geographic boundary-spanning search while considering the scarcity of the resource. However, it is largely unexplored about how the perceived competitive pressures in the industry may shape a firm's strategic location choices between local search and geographic boundary-spanning search. Furthermore, the paper also suggests that the relationships between the perceived competition and location choices of external knowledge search are contingent upon the level of tacitness in a firm's knowledge base. Prior research has demonstrated that tacit knowledge is difficult to be imitated by the competitors and transferred smoothly among actors (Kogut and Zander 1992; Kikoski and Kikoski 2004). Therefore, we argue that the tacitness of knowledge base may influence the extent to which a firm will be threatened by the competition, then increase or hinder the motivation to acquire heterogeneous knowledge and control its own valuable knowledge away from leak-out (Visser and Boschma 2004).

To test our hypotheses, our study focuses on the empirical setting on the industrial clusters in the Zhejiang province of China. Recent research focuses on the industrial cluster's role as a geographically agglomerated knowledge system (Guo, Jin, and Li 2019), and how the cluster firms interact with both intra-cluster and extra-cluster knowledge sources in the innovation process (Wang 2015). Based on a random sample of 310 cluster firms, the results show that the perceived competition has a significant and positive impact on the propensity towards geographic boundary-spanning search (referred to as 'PTGBS'). Moreover, we also find that knowledge base tacitness undermines the positive relationship between perceived competition and PTGBS. Thus, it shows that a firm's strategic location choices of external knowledge search not only depend on its competition pressure in the external environment but also on the characteristic of the knowledge base. Our study aims to make several contributions to the extant literature on industry competition and innovation search. We further advance the relationship between industry competition and innovation search literature not only by highlighting the importance of competition as a determinant of external knowledge search (Drechsler and Natter 2012) but also by pointing to the different effects that exerts on local search and geographic boundary-spanning search in terms of the propensity of search within or beyond geographical boundary of the cluster. Such findings have an important implication that competition not only influences the magnitude of external knowledge search but also the trade-off between two types of geographical knowledge search. Hence, this study provides a more nuanced view of how the interplay of industry competition and knowledge base tacitness may shape the location choices of external knowledge search.

2. Theory and hypotheses

2.1. *The trade-off between knowledge search within and beyond geographical boundaries of the cluster*

To sustain a competitive advantage, a firm need to constantly engage external knowledge search in response to the challenge of the dynamic market and intense rivalry (Laursen and Salter 2006). Scholars in the industrial cluster field have identified two strategic location choices for external search: local search and geographic boundary-spanning search (Caragliu and Nijkamp 2015). A local search involves searching for new knowledge or solutions from intra-cluster knowledge sources within the cluster's geographic boundary (Caragliu and Nijkamp 2015). The industrial cluster and economic geography literatures emphasise the superiority and dominant tendency to conduct local search which is also recognised as the most efficient mode of external knowledge search (March 1991). This is due, largely, to the greater ease and relatively lower cost of searching locally because of proximity arguments (Maskell and Malmberg 1999; Letaifa and Rabeau 2013). However, the mere reliance on intra-cluster knowledge can result in 'knowledge homogenisation' of the cluster that remains locked into an increasingly obsolete technological trajectory (Levinthal and March 1993). Furthermore, the intra-cluster knowledge sources are not only providers of new knowledge but also channels by which valuable knowledge will be spilt to rivals (Alcácer and Chung 2007). The risk is even higher when the cluster firms are searching within the industrial cluster where the local actors have more advantages in assimilating and transfer the spilt knowledge by face-to-face interactions. Therefore, it implies that the proximity of open innovators has the downside of knowledge homogenisation and negative externalities caused by overly local search.

Geographic boundary-spanning search involves searching for the new knowledge or solutions from extra-cluster knowledge sources beyond the geographic boundary of the cluster (Wang 2015). Recent research has suggested that the primary motivation for the geographic boundary-spanning search is the need to secure heterogeneous knowledge necessary to support the innovation activities of the firm (Visser and Boschma 2004). It may allow cluster firms to access more technological opportunities and acquire new knowledge that are not available through intra-cluster knowledge search (Rosenkopf and Nerkar 2001). Moreover, geographic boundary-spanning search can reduce the risk of unintentional knowledge spillover to local actors who are the major rivals competing for the scarce resources in the same region (Wang 2015). Both of geographic distance and unfamiliar contexts inhibit the unintentional spillover of knowledge to the actors outside of the industrial cluster. However, establishing and maintaining the relations with distant firms requires a large amount of time, efforts and specific investment (Bathelt, Malmberg, and Maskell 2004). It involves intense efforts and investments to develop a shared interpretative and institutional context for joint problem-solving and learning.

As the firm may have limited time and resources, there are limits to the number of external knowledge sources that a firm can manage simultaneously (Laursen and Salter 2006). The question thus becomes whether intra-cluster knowledge sources or extra-cluster knowledge sources should be prioritised. The inherent trade-off that exists between the search within and outside of industrial cluster results from the conflicting effects of local or non-local knowledge sources on the innovation. Local search helps to keep the costs of coordination and difficult of knowledge transfer at a minimum (Alcácer and Chung 2007). While critical, such local search can not provide a cluster firm of diverse and non-redundant knowledge, and also is more likely to leak out its valuable knowledge to external actors (Roper, Love, and Bonner 2017; Nan, Liu, and Ma 2018). On the other hand, geographic boundary-spanning search may enable to access the diverse knowledge distributed widely across large geographic areas (Visser and Boschma 2004). The trade-off is that it will not only spend much more resources and managerial attention, but also it will be more risky and uncertain in the firm's knowledge search process (Phene, Lindquist, and Marsh 2006). Thus, the cluster firms have to prioritise the choice between attaining proximate-based efficiency of the knowledge search process through local

search and heterogeneous knowledge through geographic boundary-spanning search. When the priority for diverse knowledge is greater than the efficiency of the knowledge search process, a cluster firm will choose to search external knowledge more widely from extra-cluster knowledge sources than from intra-cluster knowledge sources, and vice versa.

2.2. Competition and the propensity towards geographic knowledge search

Competition is regarded as a common threat related to the firm survival and may have impact on the motivation of search and strategic change (March 1991). Within an industrial cluster, competition is intense because the cluster firms usually sell seriously homogeneous products to the same market and access to the similar technologies or knowledge (Guo, Jin, and Li 2019; Ferras-Hernandez and Nylund 2019). We argue that fierce competition may trigger cluster firms to find novel ways or heterogeneous knowledge to maintain their competitive advantage (Tsai and Hsu 2014). Prior studies have suggested intensified competition may encourage competitive imitation within the cluster (Porter 2000). When it occurs, due to the fast diffusion of knowledge afforded by geographic proximity, fierce competition may result in the problem of seriously homogeneous products and then diminish a cluster firm's competitive advantage (Grant 1996; Porter 2000). To sustain the competitive advantage, the cluster firms tend to search externally to look for the superior alternatives to their current methods in both product and process, explore new markets, and find new and better ways to compete (Chen 2008; Theeke 2016). Although the intra-cluster knowledge has an economic effect for a cluster firm, it fails to provide competitive advantages due to the homogenisation or redundancy of the knowledge (Visser and Boschma 2004; Bathelt, Malmberg, and Maskell 2004). However, extra-cluster knowledge sources across locations will have a greater amount of heterogeneous knowledge due to the differences in initial endowments, the actions of actors engaged in R&D, and educational systems and regulation (Alcácer and Chung 2007). As a result, while competition is intensified, a cluster firm will be more likely to search beyond the geographic boundary of the industrial cluster to establish a differentiation advantage.

Moreover, competitive threats may raise the motivations of cluster firms for protecting their knowledge when they engage in external collaboration and knowledge sourcing (Guo, Jin, and Li 2019). Research has shown that there are substantial risks of being imitated or copied while a firm is involved in accessing external sources of knowledge (Drechsler and Natter 2012; Giarratana and Mariani 2014). The empirical study of Drechsler and Natter (2012) also finds that competitive threats decrease the probability of firms being open because a high risk exists for other firms to easily imitate a firm's products. The fact is that to obtain external knowledge, a firm has to reveal some parts of its knowledge to the external partner (Alcácer and Chung 2007). While a cluster firm spills valuable knowledge to its competitors, the competitive advantage will be eroded. Therefore, the cluster firms need to be open to engage with a broad set of external knowledge sources but also have to protect their own knowledge from being copied by its competitors (Guo, Jin, and Li 2019). Especially when the competition is becoming intense, the cluster firms will be more concerned with the risk of losing 'corporate crown jewels' involved in external knowledge searching (Drechsler and Natter 2012). Scholars find that local search will increase the risk of knowledge leak-out because co-location shortens the process and speed of knowledge spillover to competitors (Alcácer and Chung 2007). On the contrary, searching for knowledge from geographically distant sources to a large extent decreases the odds of knowledge leak-out. In sum, we propose that competition will increase the motivation for acquiring heterogeneous knowledge and protecting internal knowledge from the technology leak-out, as a result, it may compel a cluster firm to search more widely from extra-cluster knowledge sources rather than from intra-cluster knowledge sources.

H1. Competition intensity has a positive effect on PTGBS.

2.3. The moderating effect of knowledge base tacitness

The distinction between codified and tacit knowledge is the key to understanding organisational knowledge (Hsiao, Chen, and Choi 2017). Codified knowledge is always embodied in the written language, code or formula and as a consequence it can be verbalised and communicated, processed, transmitted and stored relatively easily between individuals and organisations (Zander and Kogut 1995). Tacit knowledge is deeply rooted in experience, action, procedures, skills and know-how, etc, and yet is difficult to articulate in a way that is meaningful and complete (Kogut and Zander 1992). Some scholars have identified that the knowledge can be classified on a continuum between fully tacit and fully codified (Pérez-Luño, Alegre and Valle-Cabrera 2019), so we will refer to knowledge tacitness instead of tacit or codified knowledge.

We argue that competition is less likely to increase PTGBS of those cluster firms that have higher levels of knowledge base tacitness. As noted by Kikoski and Kikoski (2004), tacit knowledge may create a learning curve for other firms difficult and time-consuming to follow. Hence, cluster firms with higher knowledge base tacitness are harder or more costly for competitors to imitate and expropriate (Kogut and Zander 1992; Kikoski and Kikoski 2004). As the distinctiveness and value of tacit knowledge assets are eroded relatively slowly, they are more likely to sustain unique capabilities and differential in products. It may be less threatened by the fierce competition and as a result will be less triggered to select the risky and costly geographic boundary-spanning search for new knowledge. In contrast, a firm with lower levels of knowledge base tacitness is more vulnerable to competition within the cluster. As codified knowledge can be easily diffused and known by other cluster firms, it is harder for a cluster firm to create unique capabilities depending on the production and use of codified knowledge (Maskell and Malmberg 1999; Hartono and Sheng 2016). As a response, while these cluster firms face intensified competition, they are more prone to emphasise on the priority for searching heterogeneous knowledge outside of the cluster to sustain the competitive position.

As previously noted, competition threat may raise more concerns on knowledge leak-out issues than search efficiency when engaging in external knowledge sourcing (Guo, Jin, and Li 2019). However, to what extent that a cluster firm responds to such a threat depends on how easily its collaborators can expropriate the focal firm's valuable knowledge base in the knowledge exchange process. Tacit knowledge is highly context-specific and hard to be transferred easily (Kogut and Zander 1992). So a tacit knowledge base can serve as an isolating mechanism that increases the difficulty of replication by outsiders (Szulanski and Jensen 2006). Because of these difficulties, while such cluster firms interact with external knowledge sources, the unintentional knowledge leak-out is less likely. When a cluster firm is more optimistic about the knowledge leak-out issues, it is more likely to draw attention on the efficiency of the knowledge search process than safeguard of a knowledge base as a response to competition. In sum, for a cluster firm with a high tacit knowledge base, the desire to seek heterogeneous knowledge and reduce knowledge leak-out triggered by competition will be lower. Thus, we suggest the following hypotheses:

- H2.** Knowledge base tacitness will negatively moderate the main positive effect of competition intensity on a cluster firm's PTGBS: the effect will be weaker when tacitness is high and stronger when tacitness is low.

3 Research methodology

3.1. Data collection and sample

We identified a random sample from a number of industrial clusters located in the Zhejiang Province of China. For budgetary reasons, we randomly selected 600 cluster firms from six cities in the Zhejiang Province. Using a systematic random sampling design, in each of six cities we sampled 100 firms from the list of cluster firms provided by the local Economic and Trade Commissions and Statistical Bureaus. We designed our questionnaire using existing scales and used its Chinese version after a

back-translation process to avoid cultural bias. The questionnaire items were measured on a five-point Likert scale, from strongly disagree to strongly agree. We adopted a double-respondent questionnaire design method by dividing the questionnaire into two parts, which were required to be finished by two top managers of the same firm independently and anonymously. We identified senior managers, mid-level managers who were knowledgeable about the firm's product innovation as the respondents. The respondents were assured that the data would be kept confidential to urge them to express their true opinions. After several rounds of follow-ups, we obtained data from 338 firms, thus achieving a participation rate of 56.3%. The industry distribution of our sample revealed that most of the cluster firms were from telecommunication and electronics (11.5%), textile and clothing (24.9%), general or special equipment (21.3%), machinery and metal (18.6%), pharmaceuticals and chemicals (11.8%), and furniture (9.2%). About 72.78% were of small size with less than 300 employees, 16.86% were of medium size with between 300 and 1000 employees and 7.40% were of large size with more than 1000 employees. About 21.60% had been in business between 5 and 10 years, with another 52.37% aged between 10 and 20 years, 10.95% aged between 20 and 30, 2.66% older, 8.28% aged less than 5 years, and 4.14% of the sample (14 firms) had the missing value. We then eliminated 28 copies of questionnaires with excessive missing data and with a single respondent. Our final sample included 310 valid questionnaires, representing an effective participation rate of 51%.

3.2. Variable measures

3.2.1. Dependent variable

We included eight external knowledge sources used in innovation: customers, competitors, suppliers of equipment, suppliers of material or components, consultants, commercial laboratories, universities or other higher education institutes, public research institutes (Laursen and Salter 2006). First, we followed Wu, Wang and Li (2015)'s study in defining intra-cluster and extra-cluster sources of knowledge. Intra-cluster refers to the municipal district (namely, the city) where the focal cluster firm is located in, and extra-cluster refers to the space outside the municipal district.

Furthermore, we asked the respondents to indicate whether their firms had used eight types of both intra-cluster and extra-cluster sources of knowledge over the last three years using a binary variable coded 1 if the source was used and 0 otherwise. Third, we summed up these codings to obtain a composite measure of intra-cluster knowledge search and extra-cluster knowledge search. Finally, we calculated the ratio of geographic boundary-spanning search to the sum of both local and geographic boundary-spanning search to measure PTGBS. Firms with a higher ratio were considered as placing a heavy emphasis on geographic boundary-spanning search, reflecting the firm's propensity to search more widely from extra-cluster knowledge sources rather than intra-cluster knowledge sources.

3.2.2. Independent and moderating variables

Following Jaworski and Kohli (1993), we used the perception of the competitive environment to measure competition intensity which includes three items. Those items reflect the managers' perception of competition that eventually counts for the decision-making in whether a firm will invest in differentiating and retaining the competitive advantages.

When the firm's knowledge base is difficult to articulate, communicate and understand, that knowledge base is considered more tacit and less codified. Based on these concepts, we adopted three fine-grained items to measure the degree of knowledge base tacitness (Zander and Kogut 1995).

3.2.3. Control variables

To control for internal R&D, we included a dummy variable taking the value of one if the cluster firm had R&D centre or department and of zero otherwise. Firm age was measured as the number of years

since the first operation in its logged format. Similarly, the firm size was measured by the natural logarithm of the number of full-time employees. Moreover, because foreign ownership and export may help the cluster firm identify useful knowledge in other regions or countries (Lyles and Salk 1996), foreign ownership was included and measured by a dummy variable that equals to 1 if the cluster firm was classified as a foreign-invested company or JVs, and 0 to indicate otherwise. We included a dummy variable to control whether the cluster firm had export activities or not. To control for differences in the financial resources, we used the adapted scale developed by Tan and Peng (2003) to measure financial slack. We introduced the construct of technological turbulence to account for the speed of change and instability in a technological environment (Jaworski and Kohli 1993). Finally, we included industry dummies to account for potential industrial differences.

3.3. Construct reliability and validity

Table 1 shows the Cronbach's α coefficients all of which were above the recommended threshold of 0.70, indicating internal consistency and good reliability of the factors generated. We conducted a confirmatory factor analysis which showed good convergent validity for the factor specification of competition intensity, technological turbulence, knowledge base tacitness and financial slack. To demonstrate discriminant validity, in Table 2 we computed the average variance extracted values for our constructs which were ranged 0.58–0.78, exceeding the 0.50 benchmark.

We carefully examined the potential multicollinearity problems of our data. None of the absolute values of the correlation between variables was above 0.50. We also mean-centred all the variables before generating the interaction terms to reduce potential multicollinearity problems. Finally, VIFs for all individual variables were all below 2 which was far below the cut-off of 10. Therefore, we were confident that our estimates were not threatened by multicollinearity problems.

3.4. Common method bias

The use of perceptual measures for both independent and dependent variables in this paper may raise concerns of common method variance in our estimations. To safeguard against this possibility, we adopted both procedural and statistical techniques recommended by Podsakoff et al. (2003) to mitigate and detect a possible common method variance problem. Procedurally, we used two respondents rather than a single respondent from each firm for data collection. Furthermore, using the confirmatory factor analysis, we compared the measurement model without the common method factor and with the factor. After adding the common method factor, the model

Table 1. Measures for all of the multi-item constructs.

Variable name	Variable construction	Standardized factor loading
Competition intensity ($\alpha = 0.74$)	There are frequent price wars or promotion wars in the industry.	0.56***
	There is fierce competition in the industry.	0.80***
Technological turbulence ($\alpha = 0.71$)	Anything that one competitor can offer, others can match readily.	0.74***
	The technology in our industry was changing quite rapidly.	0.83***
	Technological developments were highly unpredictable.	0.55***
Knowledge base tacitness ($\alpha = 0.71$)	Technological changes provided big opportunities in our industry.	0.66***
	It is difficult to comprehensively document in manuals, formulas, blueprints or reports.	0.64***
	It is difficult to precisely communicate through written documents and manuals.	0.60***
	It is difficult to comprehensively understand from written documents and manuals if without personal contact and talking to skilled employees.	0.75***
Financial slack ($\alpha = 0.81$)	The firm's retained earnings have been sufficient for market expansion.	0.67***
	It has a pool of financial resources that can be used on a discretionary basis.	0.81***
	It is able to secure necessary bank loans.	0.80***

$\chi^2 = 66.6$, $p = 0.00$; $\chi^2/\text{df} = 1.4$; CFI = 0.98; IFI = 0.98; RFI = 0.90; TLI = 0.97; NFI = 0.94; RMSEA = 0.03.

Table 2. Descriptive statistics and correlations ($N = 310$).

Variables	Mean	S.D.	1	2	3	4	5	6	7	8	9	10
1. PTGBS	0.311	0.314										
2. Firm size	5.042	1.174	−0.051									
3. Technological uncertainty	3.168	0.794	0.232***	0.009		(0.62)						
4. Firm age	2.745	0.862	0.020	0.076	0.258***							
5. Financial slack	3.329	0.786	0.146**	−0.163**	0.120*	−0.131*	(0.78)					
6. Export	0.556	0.498	0.015	0.323***	0.002	−0.028	−0.001					
7. Foreign ownership	0.195	0.397	−0.024	0.131*	−0.032	0.060	0.093 ⁺	0.140*				
8. R&D centre	0.491	0.501	0.183***	0.319***	0.177**	−0.0020	0.118*	0.032	0.113*			
9. Competition intensity	3.598	0.784	0.098 ⁺	0.167**	0.247***	0.100 ⁺	−0.003	0.195***	−0.017	−0.037	(0.63)	
10. Knowledge base tacitness	3.319	0.583	−0.112*	0.021	0.092 ⁺	0.068	−0.007	−0.031	−0.005	0.010 ⁺	0.122*	(0.58)

Note: AVE in parenthesis. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; ⁺ $p < 0.10$.

Table 3. Results of Tobit models for the effect of competition on PTGBS.

Variables	Model 1	Model 2	Model 3	Model 4
Firm size	−0.0430 ⁺ (0.0246)	−0.0473 ⁺ (0.0244)	−0.0509* (0.0238)	−0.0458 ⁺ (0.0237)
Technological uncertainty	0.117** (0.0444)	0.0980* (0.0447)	0.0948* (0.0434)	0.0985* (0.0431)
Firm age	−0.0289 (0.0416)	−0.0299 (0.0412)	−0.0186 (0.0402)	−0.0271 (0.0400)
Financial slack	0.0377 (0.0333)	0.0362 (0.0329)	0.0353 (0.0321)	0.0310 (0.0318)
Export	0.101 ⁺ (0.0548)	0.0828 (0.0548)	0.0672 (0.0534)	0.0670 (0.0529)
Foreign ownership	−0.0742 (0.0653)	−0.0594 (0.0649)	−0.0503 (0.0632)	−0.0400 (0.0628)
R&D centre	0.158** (0.0580)	0.167** (0.0575)	0.184** (0.0563)	0.173** (0.0559)
Competition intensity		0.0795* (0.0362)	0.104** (0.0359)	0.0923* (0.0358)
Knowledge base tacitness			−0.153*** (0.0438)	−0.153*** (0.0433)
Competition intensity * Knowledge base tacitness				−0.127* (0.0619)
Constant	0.136* (0.0647)	0.124 ⁺ (0.0642)	0.120 ⁺ (0.0624)	0.127* (0.0619)
Industry dummy	Included	Included	Included	Included
No. of obs	310	310	310	310
No. of left-censored obs	97	97	97	97
No. of right-censored obs	21	21	21	21
Pseudo R^2	0.078	0.088	0.115	0.124
LR χ^2	35.78***	40.58***	52.67***	56.85***
Log likelihood	−211.47097	−209.06982	−203.02439	−200.93517

Note: Standard error in parenthesis. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; ⁺ $p < 0.10$.

with a common method factor did not fit better than the model without a common method factor. These findings suggest that common method bias was not a significant problem.

4. Results

Our dependent variable is the fraction of geographic boundary-spanning search from the sum of both local and geographic boundary-spanning search, which assumes values between 0 and 1. It is ‘censored’ or has a corner solution of zero, so we perform multiple Tobit regressions analysis with Stata 12. Table 2 shows the descriptive statistics and the correlation matrix between the variables included in the empirical analysis. Table 3 reports the results for the main and moderating effects. Model 1 includes the control variables only. Model 2 introduces the main effect of competition intensity, and models 3 and 4 add moderating variables and the interactive effect of competition intensity and knowledge base tacitness. Finally, Model 5 is the full model, which is used to test the hypotheses.

In model 2, as hypothesised, we find a positive and significant relationship between competition intensity and PTGBS ($\beta = 0.080$, $p < 0.05$). Model 4 provides further support for Hypothesis 1, in that the linear term of competition intensity is positive and significant ($\beta = 0.092$, $p < 0.05$). Hence, Hypothesis 1 is confirmed. The result indicates that perceived competition may influence the location choice of external knowledge search by triggering the cluster firm to rely relatively more on geographic boundary-spanning search rather than local search. In Model 4, we introduce the interaction term of competition intensity and knowledge base tacitness to test whether there is a moderating effect of knowledge base tacitness on PTGBS. In this model, the coefficient of the interaction term is negative and significant ($\beta = -0.127$, $p < 0.05$), resulting in strong support for Hypothesis 2. It means that knowledge base tacitness will negatively moderate the main positive effect of competition intensity on a cluster firm’s PTGBS.

5. Discussion and conclusions

Based on a questionnaire survey of 310 cluster firms in the Zhejiang Province of China, we find that perceived competition creates pressures for cluster firms to shift their focus to acquire heterogeneous knowledge and reduce technology leak-out, thereby increasing the propensity for acquiring

extra-cluster knowledge. For the cluster firms with more tacit of the knowledge base, the competition will exert a lesser positive effect on PTGBS. Our study provides a more comprehensive and nuanced picture of external knowledge search within and beyond the geographic boundary of the industrial cluster under dynamic competitive environments.

Our findings yield important theoretical contributions to the literature on innovation search and industry competition. First, they provide an interesting contribution to open innovation literature on the choice of search mode decisions. To the best of our knowledge, the majority of research studies investigate the choice of governance modes such as internal and external search (Berchicci 2013) or JVs and acquisition for external knowledge sourcing (Riccobono, Bruccoli, and Perrone 2015). In response to the call for future studies into how firms' structure R&D activities (Berchicci 2013), we particularly investigate the choice between searching external knowledge beyond and within the geographical boundary of the cluster. Previous studies have reached a consensus on the need for searching knowledge simultaneously within and beyond the geographic boundary of the industrial cluster (Wu and Wu 2014; Wu, Wang, and Li 2015). However, the common emphasis of those studies has to be given to treat local search and geographic boundary-spanning search as a dichotomy or two quite different alternatives (Wu, Wang, and Li 2015) or examine the interactive effect of two sources of knowledge search (Wu and Wu 2014). The results of Wu and Wu (2014) show that extensive local search and international search complement each other and produce a more positive impact on product innovation. Our operationalisation of the construct contributes to a more fine-grained conceptual understanding of the trade-off between two different geographical scales of knowledge sources as an organisational adaptation to the competitive environment. The managerial implication is the need for senior managers to prioritise the allocation of resources between local search and geographic boundary-spanning search and makes a trade-off in a resource-constrained context.

Second, our findings contribute in understanding the relationship between industry competition and external knowledge search. Industry competition has a critical role in determining a firm's strategy of resource allocation and acquisition (Melville, Gurbaxani, and Kraemer 2007) and search behaviour pattern (Chen and Miller 2007). So far, only limited empirical studies pay attention to the direct or contingent roles of industry competition in shaping a firm's external knowledge search behaviour. For example, García-Granero et al. (2014) find that increased competition will decrease the probability of pursuing an exploratory oriented external search strategy. Wang and Guo (2019) indicate that industrial competition will exert a negative moderating effect on the relationship between absorptive capacity and external knowledge search. Our study extends the existing studies by proposing that industry competition may shape strategic location choices of external knowledge search in the particular context of the industrial cluster. Furthermore, our results inform the ongoing debate on the choice of local search and geographic boundary-spanning search by firms under the fierce competitive environment. On the one hand, the competition will increase the incentive of cluster firms for acquiring heterogeneous knowledge from external knowledge sources. The long-term industrial competitiveness is related to the ability of firms to upgrade their knowledge base continuously (Maskell and Malmberg 1999). On the other hand, competition may raise the firm's concerns for leak-out of valuable knowledge to external actors (Alcácer and Chung 2007; Guo, Jin, and Li 2019), thereby reducing the use of external sources (Giarratana and Mariani 2014). Therefore, our findings inform this on-going debate by theoretically proposing and empirically showing that competition not only influences the magnitude of external knowledge search but also the trade-off between two different geographical orientation of the knowledge search. The managerial implication of the results may present a partial solution to such a dilemma by suggesting that the cluster firm may choose and trade-off between two different geographical scales of knowledge search according to their relative advantages.

Furthermore, our findings shed light on the role of knowledge tacitness in knowledge transfer and external search process. Scholars have reached a consensus that proximity matters in transferring tacit knowledge because such a type of knowledge exchange continues to require regular face-to-face contact and a high degree of mutual trust and understanding (Malmberg and Maskell 2002;

Nan, Liu, and Ma 2018). However, most of those studies focus their attention on the tacitness of transferred knowledge, instead of the knowledge base of the firm conducting external knowledge search (Hsiao, Chen, and Choi 2017). This study, to the best of our knowledge, is one of the first to systematically investigate how the extent to which tacit knowledge constitutes the knowledge base of the cluster firm will influence the geographic location choices of the external search as a response to the pressure of competition. Our study indicates that knowledge base tacitness weakens the positive effect of competition intensity on PTGBS by reducing the threaten by the fierce competition, while lessening the concerns for controlling knowledge leak-out issues. It then complements arguments on how knowledge attribute influences the transfer, exploitation and control of innovation-related knowledge during the innovation search process.

Our study has a number of limitations that point to directions for future research. First, given that perceptual measures were used for both the independent and moderating variables, it might be the possibility of common method bias. Although we employed both procedural and statistical precautions, the potential of this bias could not be completely ruled out. Second, the survey data were gathered at one point in time, which prevented us from studying causal relationships between competition and PTGBS. A longitudinal study may provide further insight into the dynamics of how the cluster firms trade-off between types of knowledge search to adapt competition in the market. Third, our sample was selected from the industrial clusters in China. This might raise the concern of the generalisability issue of our findings. Additional empirical research, using samples of firms coming from other countries, is warranted. Furthermore, future studies may consider the coexistence of competition and cooperation between the cluster firm. Although the cluster firms compete fiercely, it is worthwhile to examine how competition and cooperation interactively influence the location choices of external knowledge search across the geographic boundary of the industrial cluster. Finally, this paper examined the moderating effect of knowledge base tacitness, but it is reasonable to expect that other internal contingencies, such as organisational slack and organisational inertia, may also influence the relationship. Perceived competition may provide incentives for a cluster firm to search comparably more from extra-cluster knowledge sources, but it is also critical for a cluster firm to have sufficient resource and capacity to adjust their external search strategy.

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Notes on contributors

Qiang Li is an associate professor of Business School of Zhejiang University City College, Hangzhou, China. His research topics include innovation network and knowledge search in emerging economies.

Jingjing Guo is an associate professor at Institutes of Science and Development, Chinese Academy of Sciences and School of Public Policy and Management, University of Chinese Academy of Sciences. Dr. Guo's research concentrates on topics such as innovation management and innovation policy.

ORCID

Qiang Li  <http://orcid.org/0000-0002-5077-2194>

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